

Project

DS 5110: Big Data Systems

Summer 2026

Lecture 5

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Announcement

- You should expect to receive an OpenRouter API key by EOD today
- A1 is out
 - Due: 11:59pm this Friday
- Watch out for Canvas announcements for:
 - Redeeming Google Cloud credits for projects
 - Project idea selection sheet

Project milestones

- A 2-week development project
 - I'll supply ideas (and codebase for some)
 - You'll build the idea into an **MVP** (minimum viable product) with AI coding
 - May 21: Team signup
 - May 27: Picking a project idea
 - May 29: Project checkpoint 1
 - June 5: Project checkpoint 2
 - June 11+12: Project presentation
 - June 13: Project everything due
- } ← This week

Project scope and methodology

- Keep the scope practical and manageable
 - Aim for something that can be completed in 1—1.5 weeks
- Be cautious with open-ended research ideas
 - They may take longer than expected
- Use AI coding tools effectively
 - Boosting development productivity by 5—10x
- Example project directions
 - Reproduce (and extend) examples from an open-source project
 - Build a small MapReduce / Spark application and measure performance
 - Develop a focused prototype and evaluate it with clear metrics

Project ideas

- Theme 1: autoresearch
- Theme 2: Flash-MoE
- Theme 3: Evolving domain algorithms w/ OpenEvolve
- Theme 4: In-browser video understanding w/ WebGPU
- Theme 5: Large AI model compression
- Theme 6: Hugging Face ZeroGPU space
- Theme 7: Data system experimentation

- Theme X: Your own ideas?

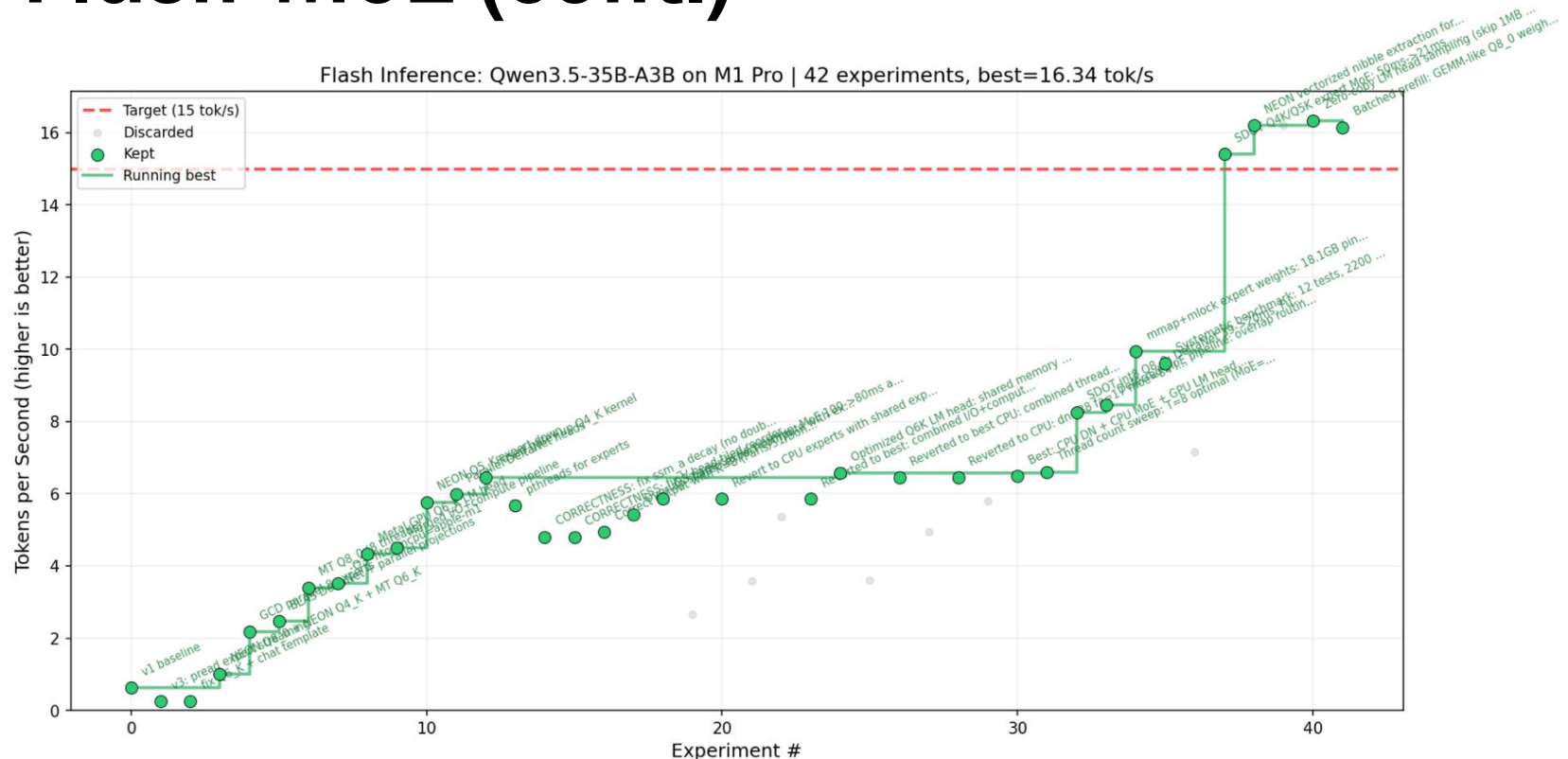
autoresearch: AI agents for autonomous experimentation

- Problem:
 - ML research is slow and manual
 - Many ideas require repeated trial-and-error
- High-level idea:
 - Give an AI agent a small training setup
 - Let it edit code, train, evaluate, and repeat
 - Keep good changes, discard bad ones
 - Review experiment logs and progress plots afterward
- Project ideas:
 - Reproduce Karpathy's autoresearch results
 - Use autoresearch for SLM (small language model) fine-tuning
 - Apply autoresearch to another domain

Flash-MoE: Running and optimizing a large MoE model on a personal laptop

- Problem:
 - Local LLM deployment is becoming increasingly popular
 - Large models often exceed laptop memory limits
 - Consumer devices need smarter memory strategies
- High-level idea:
 - Use MoE sparsity: only a few experts are active per token
 - Keep non-expert weights resident in memory and store expert weights on SSD or mem-mapped storage
 - Load selected experts on-demand during decoding

Flash-MoE (cont.)



For development: Open-weight models are generally not smart enough for such tasks. This project is better suited for commercial AI (Claude Code, Codex CLI).

Suggestion: Do not start from scratch. Instead, start from an existing open-source implementation (see GitHub repos provided in Google Doc)

Evolving domain algorithms w/ OpenEvolve

- Problem:
 - Many domain-specific algorithms are tunable
 - Small design choices or parameter selection can affect performance
- High-level idea:
 - Choose a small, clean, self-contained algorithmic task (**avoid projects that involve a large or overly complicated codebase**)
 - Define a baseline, correctness checks, and a score function
 - See examples from OpenEvolve GitHub project
- *Exploration: Compare proprietary commercial AI models vs. open-weight models*

WebGPU-based in-browser video understanding

- Problem:
 - Modern AI apps often depend on cloud GPUs and CUDA development can be heavy
 - Video understanding raises privacy concerns
- High-level idea:
 - Build a WebGPU-based AI app that runs fully in browser
 - Use Apple FastVLM for lightweight video understanding
 - Capture frames from webcam, generate captions, detect events, or answer visual questions

Large AI model compression

- Problem:
 - LLMs consume large amounts of storage
 - Fine-tuned LLMs are numerically similar to base models
 - General-purpose lossless compression often don't work
- High-level idea:
 - Exploit redundancy across related models / tensors
 - Use XOR-based delta compression for similar models / tensors
- Project ideas:
 - Build a local Hugging Face cache compressor
 - Integrate BitX into existing tools (dvc, huggingface_hub)

Hugging Face ZeroGPU spaces

- Problem:
 - GPU-backed demos are expensive to host
 - Dedicated GPUs are often idle between user requests
- High-level idea:
 - Use ZeroGPU as a serverless GPU platform for Gradio apps
- Note: This project requires a paid HF Pro account

Data system experimentation

- Comparing data systems
 - Pick two to three representative data systems (Spark, DuckDB, Lambda), implement representative algorithms (Kmeans, PageRank, Sort, etc.), and cross-compare these systems
- Scaling X on AWS Lambda
 - Where X = Some MapReduce or ML algos

Analyze your own dataset

- Pick a reasonably large dataset, analyze it, build a dashboard, and report your findings

Bring your own ideas?

- Come talk to me!
- Guiding principles
 - Align closely with **AI systems or data systems**
 - Aim for a **small project scope** that can be completed within 2 weeks

Scope distribution

- Some projects are narrower in scope while others are broader
- You may pick one or adjust the direction/scope
 - Come discuss with me
- My goal: to get a palatable distribution of the project ideas
 - Teams working on relevant ideas within same theme are encouraged to **coordinate** and make sure project scopes are distinct
- Your goal: choose a project that genuinely interests you

Last but not least...

- Ask me anything about your projects
 - Challenges in using AI tools
 - Project design, scope, and planning
 - Implementation (e.g., choice of open-source software)
- How to reach me
 - During class or office hours
 - Friday studio
 - On Ed
 - By email

I'm excited to see the amazing MVPs you'll build by the end of Week 4!